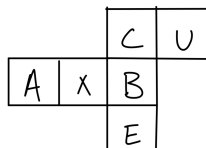


Otago Challenge 2012 Set 3

1 Problems

Problem 1.1 The net below is cut out along the outer edges and folded along the inner dotted lines to form a cube.

What letter on the cube is opposite X ?



Problem 1.2 From the coin jar at home, Blake pulled out a total of one dollar in 10 cents and 20 cent coins. There were 8 coins altogether.

How many 10 cents coins did Blake Have?

Problem 1.3 The second and third digits of the four-digit number $6dd5$ are the same. The number is exactly divisible by 9.

What digit does d represent?

Problem 1.4 Ali, Brad and Cory collect rugby cards from their favourite breakfast cereal. When counted together, Ali and Brad have 80 cards, Brad and Cory have 94 cards, and Ali and Cory have 106 cards.

How many cards do they have all together?

Problem 1.5 Five identical rectangles are placed together as shown (not to scale).

What is the area of one of these rectangles (in square centimetres?)

